

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 897+020

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

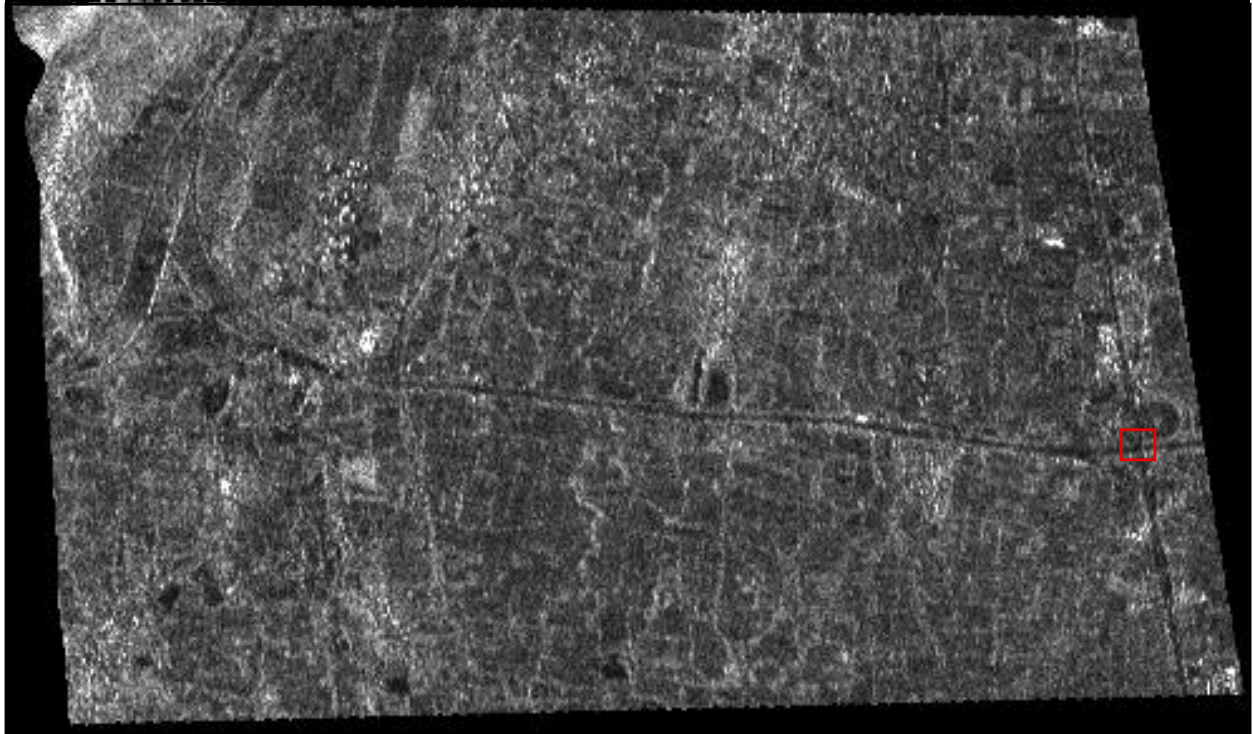
Flight Direction - DESCENDING

Polarization – VV+VH

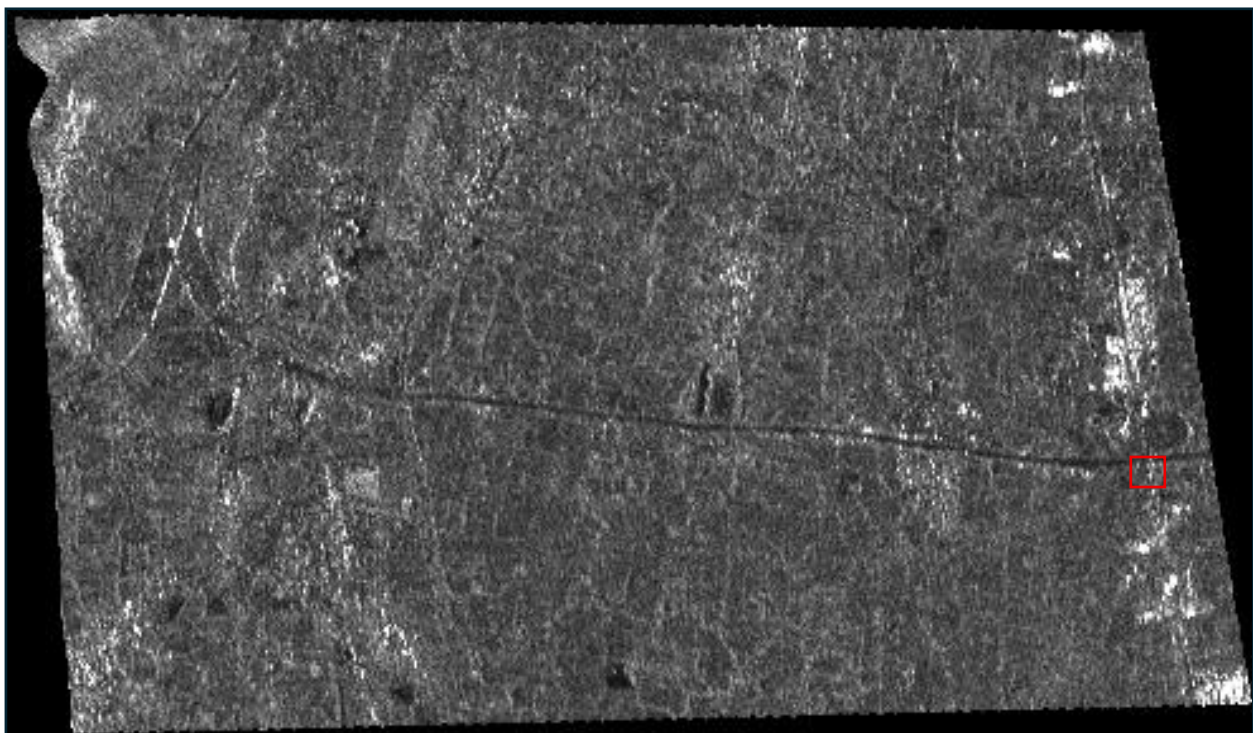
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

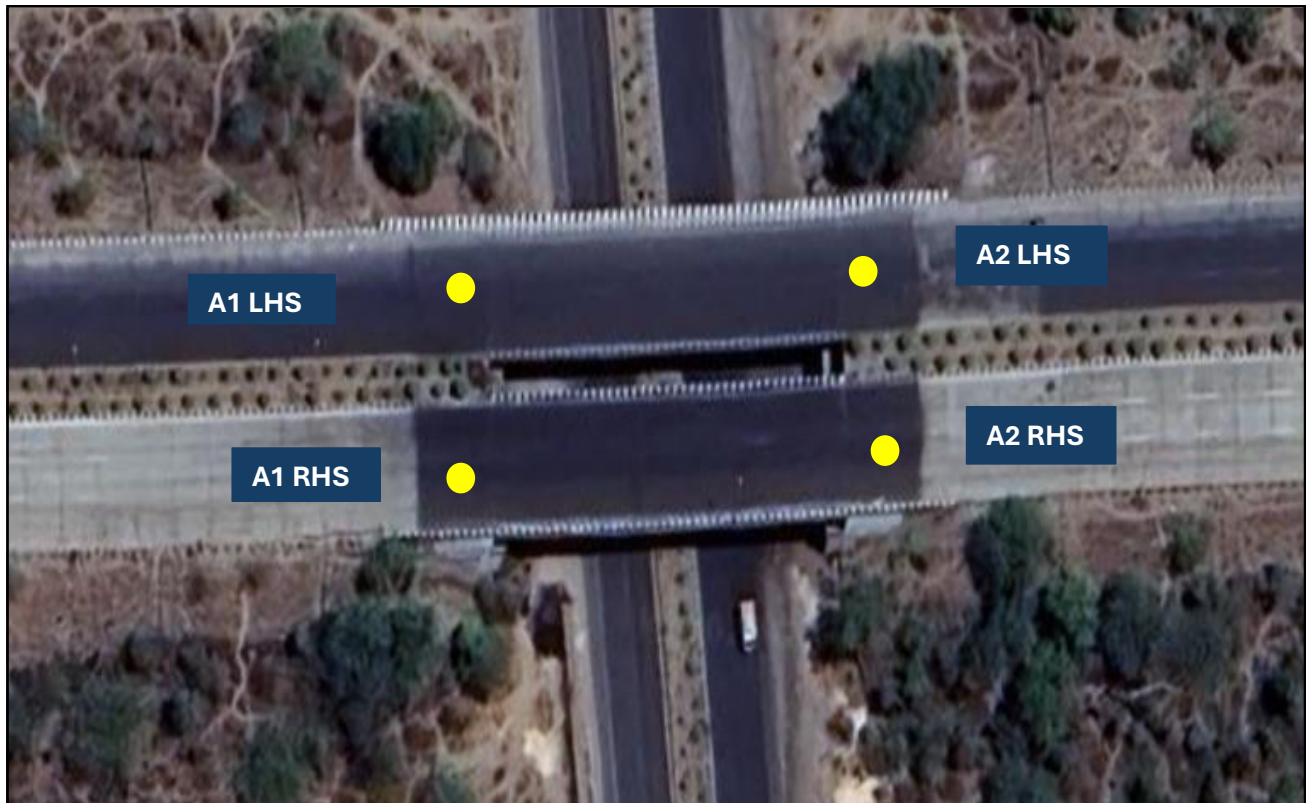


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

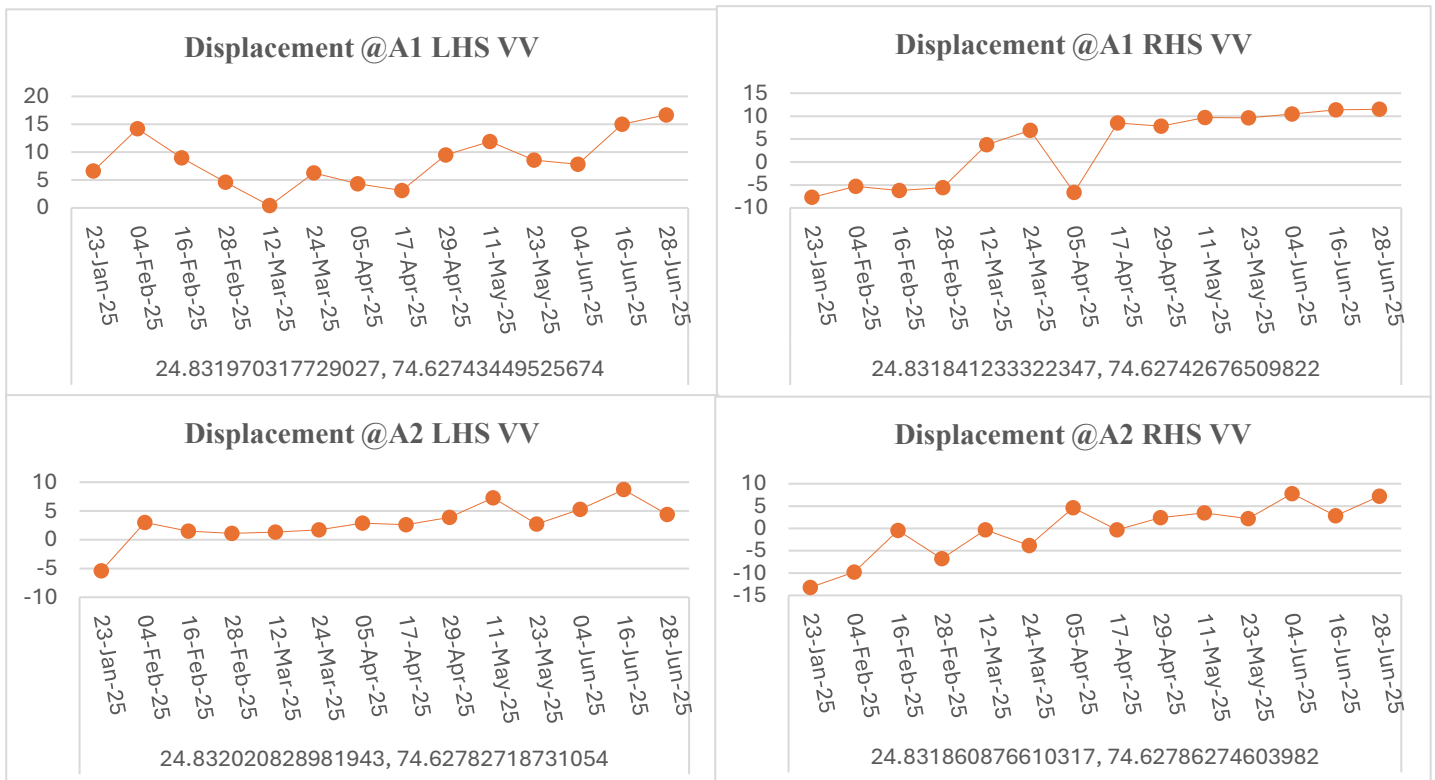
VUP – CH-897+020

Length – 59.74M

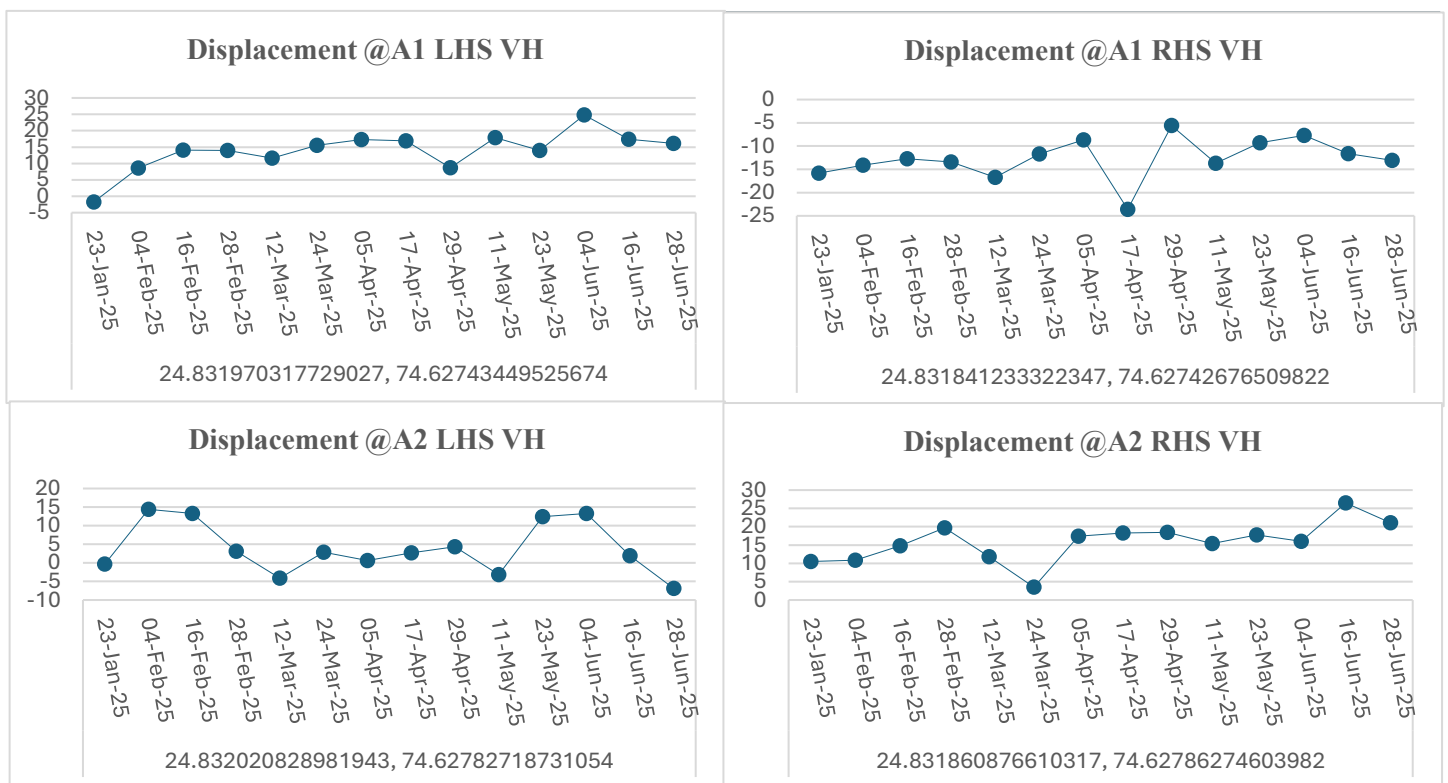


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



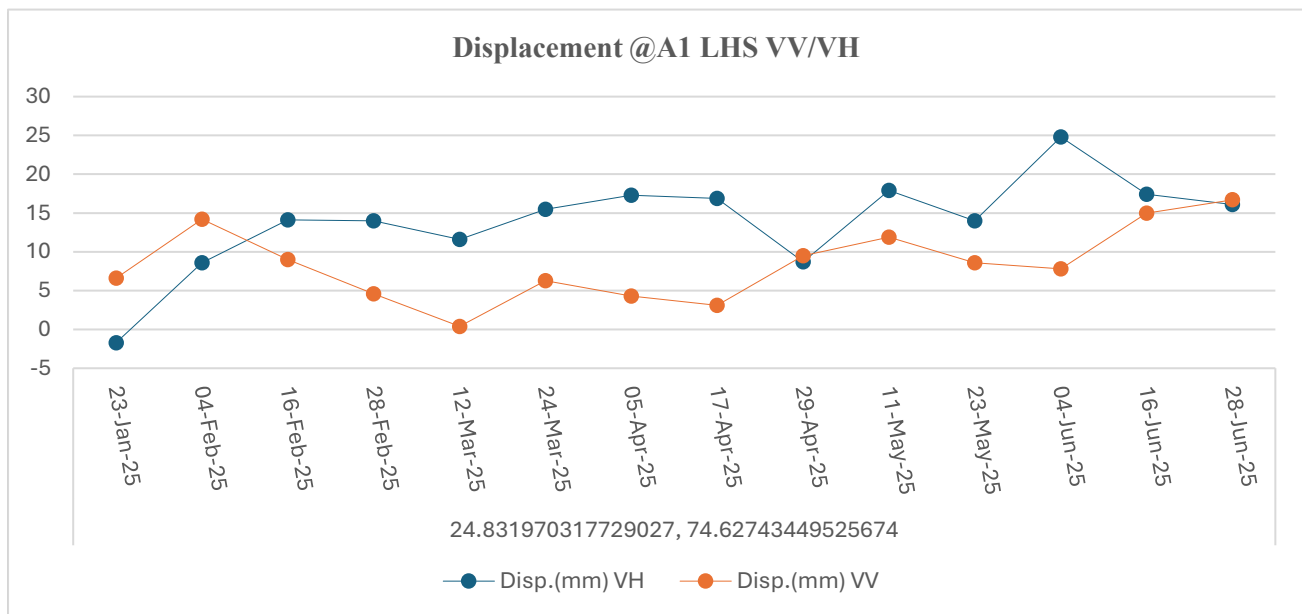
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.831970317729027, 74.62743449525674)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** exhibits moderate variability with an overall positive dominance across the monitoring period (Jan–Jun 2025), indicating net vertical uplift at the A1 LHS location.
- **Horizontal Displacement (VH)** Horizontal displacement shows high variability with a strong positive trend, ranging roughly between 0 mm and 25 mm.

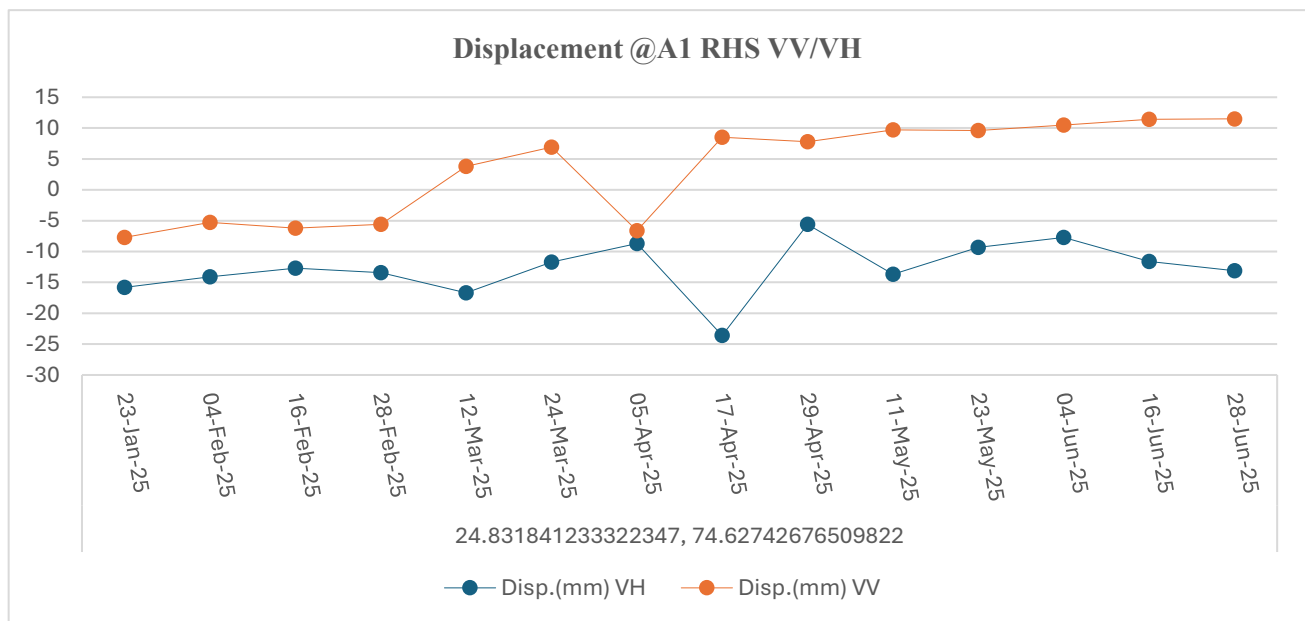
Critical Events & Anomalies:

- **February 2025:** VH rises sharply to 8–10 mm, while VV peaks near 14 mm, representing an initial coupled lateral and vertical uplift event.
- **April 2025:** VH remains elevated (17 mm), while VV increases to 9 mm, indicating renewed vertical uplift under steady lateral displacement.
- **June 2025:** VH reduces slightly to 16–17 mm, while VV increases again to 16–17 mm, confirming continued combined lateral and vertical uplift toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.831841233322347, 74.62742676509822)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows high variability with strong negative dominance, ranging roughly between –6 mm and –26 mm.
- **Vertical displacement (VV)** Vertical displacement exhibits a clear transition from early settlement to sustained uplift over the monitoring period. VV values range approximately from –7 mm to +12 mm.

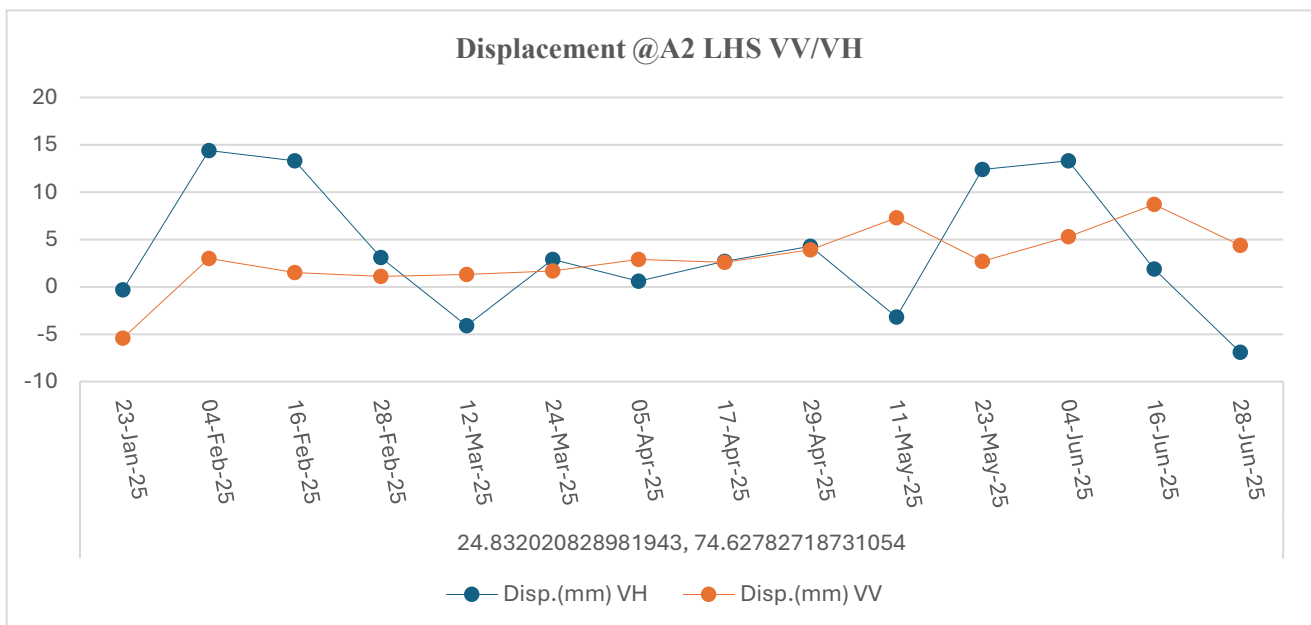
Critical Events & Anomalies:

- **March 2025:** VV rises sharply to +6 mm, while VH deteriorates to –17 mm, marking a decoupled deformation event with vertical uplift occurring under increased lateral settlement.
- **May-June 2025:** VV continues a steady upward trend, stabilising around +10–12 mm, while VH fluctuates between –10 and –15 mm, confirming sustained vertical uplift combined with ongoing lateral settlement toward the end of the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.832020828981943, 74.62782718731054)



Key Observations:

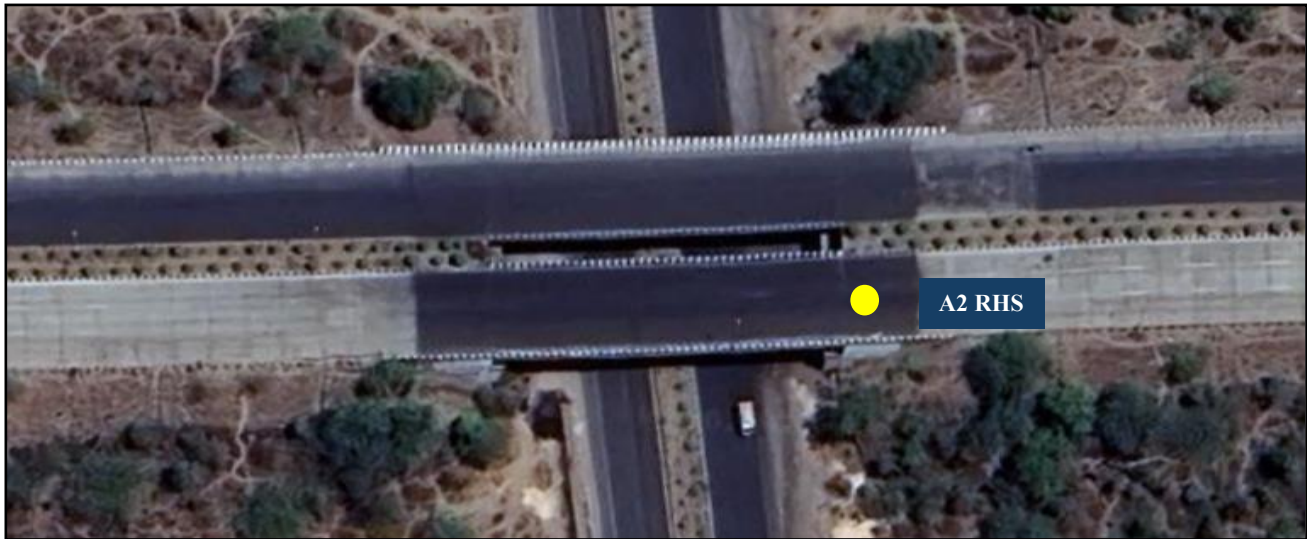
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits high variability with alternating uplift and settlement phases, ranging roughly between -7 mm and $+14$ mm.
- **Vertical Displacement (VV)** Vertical displacement shows moderate variability with a predominantly positive tendency over the monitoring period (Jan–Jun 2025), indicating overall vertical uplift at the A2 LHS location.

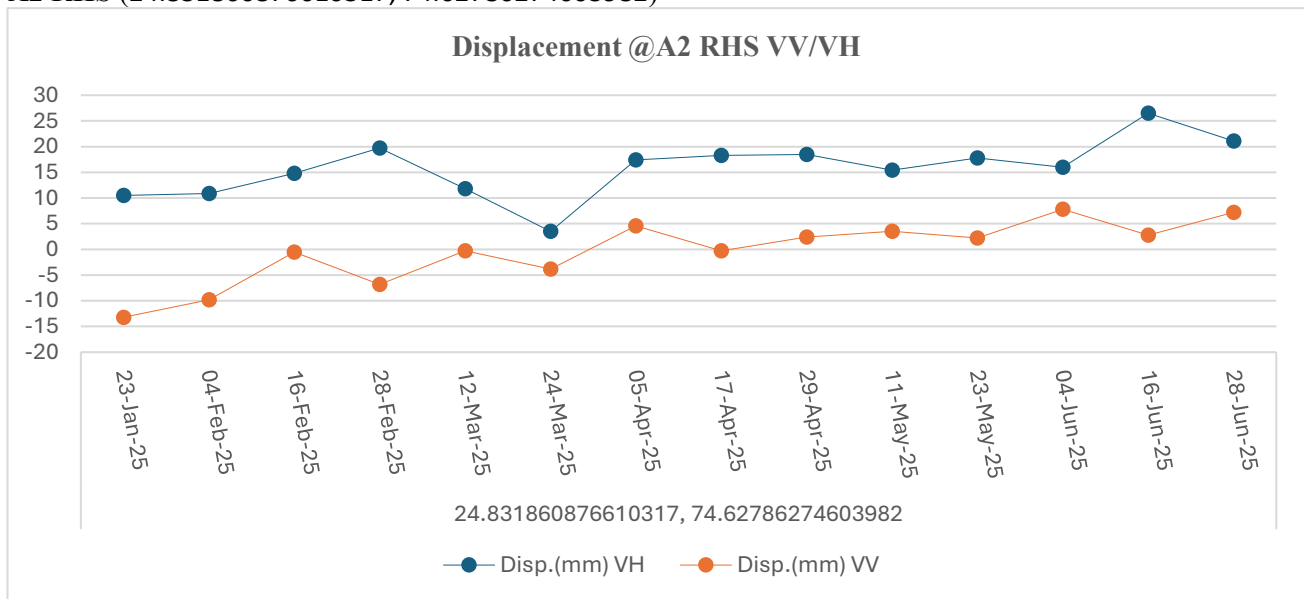
Critical Events & Anomalies:

- **February 2025:** VH rises sharply to $+14$ mm, while VV increases to $+3$ mm, marking a combined lateral and vertical uplift event.
- **June 2025:** VH drops to $+2$ mm and then to -7 mm, while VV stabilises around $+4$ mm, confirming continued lateral variability with relatively stable vertical behaviour toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.831860876610317, 74.62786274603982)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** exhibits moderate variability with a gradual transition from early settlement to mild uplift over the monitoring period values range approximately from -14 mm to +8 mm.
- **Horizontal (VH) displacement** shows high variability with strong positive dominance, ranging roughly between +6 mm and +27 mm.

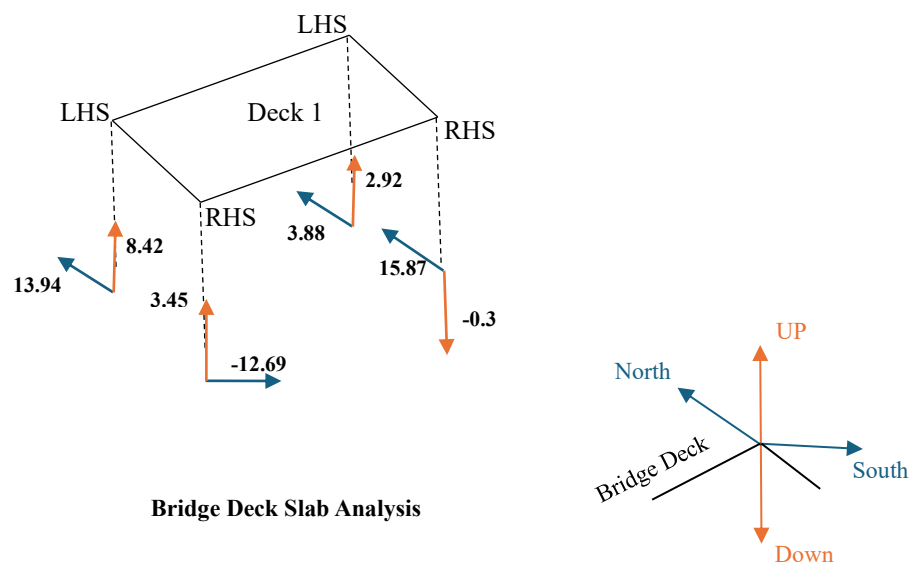
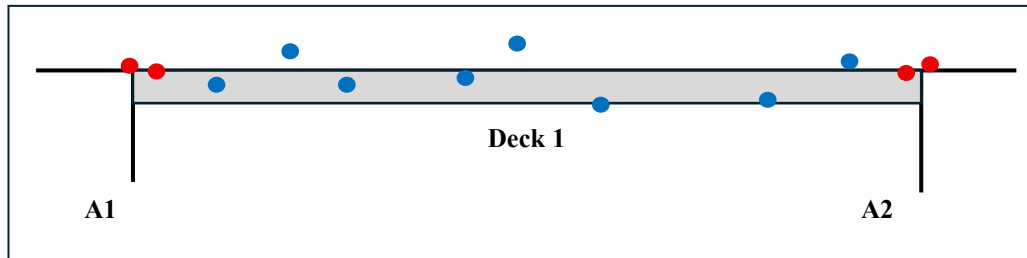
Critical Events & Anomalies:

- **February 2025:** VH increases to +12 mm, while VV improves slightly to -10 mm, marking a partial vertical recovery under continued lateral movement.
- **March 2025:** VH drops sharply to +4 mm, while VV is mildly negative, marking a temporary relaxation phase affecting both components.
- **June 2025:** VH reaches one of its highest values, while VV peaks near +8 mm, representing a strong combined lateral and vertical uplift event.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab continues to exhibit non-uniform vertical displacement, indicating differential response across LHS and RHS supports. The reduced magnitude of settlement compared to uplift highlights an overall uplift-dominated vertical response with localized vertical imbalance.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates clear asymmetry across the deck width, reflecting torsional and lateral deformation effects. The largest horizontal displacement occurs at the outer RHS region, reaching approximately +15.87 mm, indicating significant lateral movement toward the defined direction.

CONCLUSION:

- The combined vertical and horizontal displacement vectors indicate a non-symmetric and torsion-influenced deformation pattern. The bridge deck slab exhibits asymmetric deformation with torsional characteristics, marked by maximum vertical uplift at the LHS (+8.42 mm) and peak horizontal displacement at the RHS outer region (+15.87 mm).